Trial SET #8 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

1.

A new computer does a calculations in b hours, and an old computer does c calculations in d minutes. If the two computers work together, how many calculations do they perform in m minutes?

(A) $m\left(\frac{ac}{60bd}\right)$

(B) $60m\left(\frac{a}{b}+\frac{c}{d}\right)$

(C) $m\left(\frac{60a}{b}+\frac{c}{d}\right)$

(D) $m\left(\frac{a}{b}+\frac{c}{d}\right)$

(E) $m\left(\frac{a}{60b}+\frac{c}{d}\right)$

2.

Which of the following lines is perpendicular to the line $y=\frac{3}{2}x+7$?

(A) $y=\frac{3}{2}x-7$

(B) $y=-\frac{3}{2}x+7$

(C) $y=\frac{2}{3}x+7$

(D) $y=-\frac{2}{3}x+7$

(E) $y=\frac{3}{2}x+\frac{1}{7}$

3.

If $\sec\theta=2$, then $\cos\theta\sec\theta=$

(A) 2

(B) 4

(C) $\frac{1}{4}$

(D) 1

(E) 0

4.

What is the range of $f(x) = \sqrt{4 - x^2}$?

(A) $y \geq 0$
(B) $y \geq 2$
(C) $-2 \leq y \leq 2$
(D) $0 \leq y \leq 2$
(E) $y \leq 2$

5.

The graph of $x^2 - xy = 4$ has which of the following symmetries?

(A) Symmetric with respect to the x-axis
(B) Symmetric with respect to the y-axis
(C) Symmetric with respect to the origin
(D) Symmetric with respect to both axes
(E) None

6.

Cost is a function of the number of units produced as given by: $C(n) = 0.01n^2 - 90n + 25{,}000$. How many units, n, produce a minimum cost C?

(A) 500
(B) 4,500
(C) 9,000
(D) 18,000
(E) −177,500

7.

$\sqrt[5]{\sqrt[4]{\sqrt[3]{\sqrt{n}}}} =$

(A) $n^{\frac{1}{17}}$
(B) $n^{\frac{1}{19}}$
(C) $n^{\frac{1}{60}}$
(D) $n^{\frac{1}{120}}$
(E) $n^{\frac{77}{60}}$

8.

What is the maximum value of the function $f(x) = \frac{1}{x}$ over the interval $\frac{1}{2} \leq x \leq \frac{3}{2}$?

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) 2

(D) $\frac{2}{3}$

(E) infinity

9.

If $f(x) = 4^{\frac{x}{4}}$, then $f\left(\log_2 \frac{1}{256}\right) =$

(A) $\frac{1}{65,536}$

(B) $\frac{1}{64}$

(C) $\frac{1}{16}$

(D) $3\sqrt{3}$

(E) 16

10.

The operation Φ is defined for all real numbers a and b by the equation: $a \; \Phi \; b = a^{-b} - 3b$

If $n \; \Phi \; -2 = 70$, which of the following could equal n?

(A) 7

(B) 9

(C) −8

(D) 8.7

(E) $\frac{1}{8}$

11.

In Figure 6, the length of each edge of the cube is s. If A and B are midpoints of two edges, what is the perimeter of ABCD?

(A) $4s\sqrt{5}$

(B) $2s + s\sqrt{\frac{5}{2}}$

(C) $2s + s\sqrt{5}$

(D) $3s$

(E) $4s$

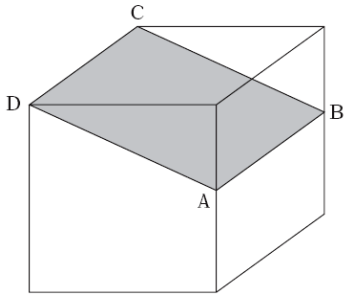


Figure 6

12.

The probability that it will snow tomorrow is $\frac{2}{3}$ and, independently, the probability that it will also snow the day after tomorrow is $\frac{1}{5}$. What is the probability that it will snow tomorrow but *not* the day after tomorrow?

(A) $\frac{2}{15}$

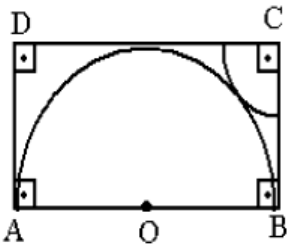
(B) $\frac{2}{5}$

(C) $\frac{3}{4}$

(D) $\frac{8}{15}$

(E) $\frac{22}{15}$

13. In the given figure, ABCD is a rectangle, a semicircle centered at O and a quarter circle centered at C. If $|AD| = 5$ cm then find the radius of small circle.



A) $5\sqrt{2}+1$ B) $5+\sqrt{2}$ C) $5\sqrt{2}-1$

D) $5\sqrt{2}-5$ E) $5\sqrt{2}$

14.

The linear regression model $C = 13.2m + 20.5$ relates the calories burned using a new exercise machine (C) to the number of minutes a person uses the machine (m). Which of the following statements about this model must be true?

I. When a person spends 22 minutes using the machine, the predicted number of calories burned is approximately 311.
II. There is relatively no correlation between C and m.
III. A person burns approximately 13.2 calories each minute he or she uses the machine.

(A) I only
(B) III only
(C) I and II only
(D) I and III only
(E) I, II, and III

15.

What is the middle term of the expansion of $(x + 2)^6$?

(A) $15x^4$
(B) $20x^3$
(C) $60x^4$
(D) $160x^3$
(E) $240x^3$

16.

When x is increased by 5, the result is 60% of increasing x by 15. What is the value of x?

A. 6
B. 8
C. 10
D. 12
E. 14

17.

Some number c is added to the three numbers 1, 13, and 61 to create the first three terms of a geometric sequence. What is the value of c?

(A) 1
(B) $\frac{1}{2}$
(C) 2
(D) 3
(E) 5

18.

The graph of $f(x) = x^3$ is translated 6 units up, 2 units right, and reflected over the x-axis. If the resulting graph represents $g(x)$, then $g(-1) =$

(A) -21
(B) 5
(C) 14
(D) 21
(E) 33

19.

For which of the following functions is it true that $f(-2) = f(2)$?

(A) $f(x) = \frac{3}{x}$

(B) $f(x) = 2 - x$

(C) $f(x) = x + 1$

(D) $f(x) = x^2 + 1$

(E) $f(x) = 2x^3$

20.

The rectangle in Figure 6 is rotated about side WZ.
What is the volume of the resulting solid?

(A) 432
(B) 108π
(C) 432π
(D) 72π
(E) 330



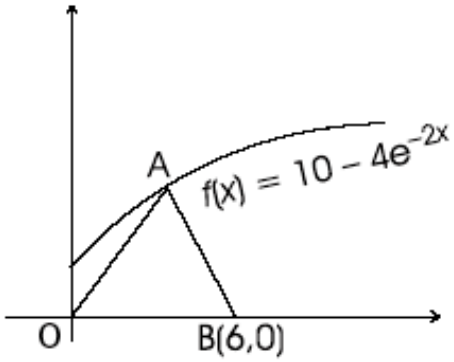
21.

A ship 55 km from a shore springs a leak which admits 2 tonnes of water in 6 min. 80 tonnes would suffice to sink it, but the pumps can throw out 12 tonnes an hour. The average rate of sailing for the ship to just reach the shore before sinking is...

(a) 6.5 kmph.
(b) 6 kmph.
(c) 5.5 kmph
(d) 8 kmph.
(e) 4 kmph.

22. (Use approximate calculations to solve the following task)

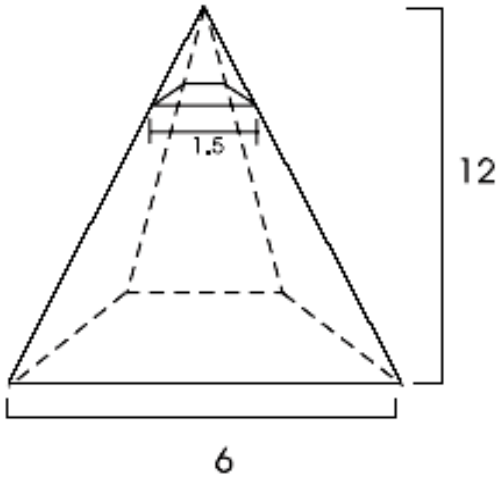
The graph of $f(x) = 10 - 4e^{-2x}$ is shown below. What is the area of triangle ABC if $OA = AB$?



(A) 25
(B) 60
(C) 45
(D) 50
(E) 30

23.

Consider a pyramid with a square base side of 6 inches and with a height of 12 inches, as shown below. If we cut off the top of the pyramid parallel to the base 3 inches from the tip, what is the volume of the remaining solid?



(A) 141.75
(B) 140
(C) 135.48
(D) 144
(E) 130

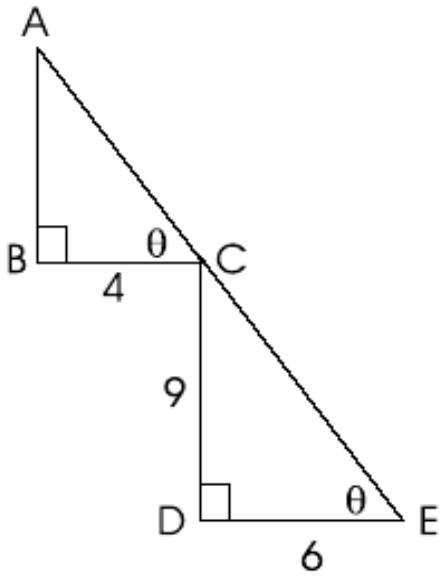
24.

In how many ways can the letters of the word SICILY be arranged using all of the letters?

(A) 60
(B) 120
(C) 240
(D) 360
(E) 720

25. (Use that $\sqrt{13}$ =3.60...)

If triangles *ABC* and *CDE* are similar and other dimensions are as shown below, what is the length of segment *AE*?



(A) 18.03
(B) 17.05
(C) 16.86
(D) 19.85
(E) 18.28

26.

The graph of $y = -x^4 + 12x - 18$

(A) intersects the x-axis at exactly one point.
(B) intersects the x-axis at exactly two points.
(C) intersects the x-axis at exactly three points.
(D) intersects the x-axis at exactly four points.
(E) does not intersect the x-axis.

27.

June 25, 1982, fell on a Friday. On which day of the week did June 25, 1987, fall? (Note: 1984 was a leap year.)

A. Sunday
B. Monday
C. Tuesday
D. Wednesday
E. Thursday

28.
Which numbers have an odd number of divisors?

A. Odd numbers
B. Perfect squares
C. Prime numbers
D. Perfect cubes
E. Positive integers

29.
What is the next number in the series below?

7 16 14 12 21 8 ___

a. 4
b. 15
c. 28
d. 24
e. 32

30.
$x^2 + (3m + 4)x + m - 1 = 0$ is given. If $x_1^2 + x_2^2 = 10$ then find one possible value of m.

A. 2 B. 4/9 C. 1 D. - 4/3 E. -2

Answers:

Right answers:

1.
2. D
3. D
4. D
5. C
6. B
7. D
8.
9.
10. C
11. C
12. D
13. D
14. D
15. D
16.
17.
18. D
19.
20. B
21. C
22. E
23. A
24. D
25. A
26. E
27. E
28. B
29. C
30. E